

Clinical Practice Guidelines for Assessment and Management of Anxiety and Panic Disorders in Emergency Setting

Nadukuru Nooka Raju, Kampalli S. V. R. Naga Pavan Kumar, Gyan Nihal¹

GVP Medical College, Visakhapatnam, Andhra Pradesh, ¹Mamata Medical College, Khammam, Telangana, India

E-mail: drnnraju@yahoo.com

Submitted: 24-Jul-2022, Revised: 23-Dec-2022, Accepted: 28-Dec-2022, Published: 30-Jan-2023

INTRODUCTION

Anxiety disorders, primarily characterized by pathological anxiety, are very common in the general population with estimates ranging from 20 to 30 percent, and the lifetime prevalence rate is calculated at 16.6 percent.^[1] It is to be noted that pathological anxiety is experienced as more intense and exaggerated which arises without any real threat.

The word anxiety was taken from the Latin root “anxieta” meaning disturbance in the mind about an uncertain event, and the Greek root “anxo” meaning to squeeze, strangle, or press tight.^[2]

Anxiety is an unpleasant emotional state that is associated with psychophysiological changes as a result of an intrapsychic conflict. Anxiety disorder is a disorder in which the most prominent disturbance is anxiety or in which patients experience anxiety when they refuse to give in to their symptoms.^[3]

Panic is an acute, intense attack of anxiety associated with disorganization in personality. It is characterized by overwhelming anxiety and feelings of impending doom. A panic attack is an acute intense anxiety episode, which occurs in panic disorder (PD), major depression, schizophrenia, and somatization disorder. PD is characterized by acute intense anxiety attacks in the presence or absence of agoraphobia.^[3]

The onset of panic symptoms is often spontaneous and rapid. Many patients seek help in an emergency department (ED) because of these characteristics, combined with its overwhelming intensity.^[4]

Anxiety-related complaints are commonly associated with alcohol and substance abuse, which further complicates the emergency physician’s assessment.^[5]

PDs and exacerbations of previous acute stress symptoms are common patient presentations in the ED for primary anxiety disorders. Individuals with anxiety disorders

may feel debilitated during an anxiety attack and feel embarrassed after the episode. To support the patient both during and after the anxiety episode, it is essential for ED clinicians to be aware of anxiety and PDs.^[6] Table 1 outlines clinical predictors of anxiety caused by an underlying medical disorder.

CLINICAL PRESENTATION

In ED, anxiety presentations may be classified into four groups^[5]:

- Primary psychiatric illness, such as generalized anxiety disorder
- Response to a stress or stressful event, such as acute stress disorder
- Medical conditions or substance abuse that mimic the symptoms of anxiety, such as hyperthyroidism
- Anxiety disorder comorbid with another physical or mental illness

Six stages of development have been identified for PD, with stage 1 or stage 2 being the most common for patients. These stages are (1) limited symptom attack, (2) full panic attack, (3) hypochondriasis, (4) limited phobic avoidance, (5) extensive phobic avoidance, and (6) secondary depression. Stage progression is associated with increasing degrees of disability and corresponding treatment implications.^[8] It is believed that the disorder is less likely to worsen if the diagnosis is made in stage 1 or 2 and treated.^[9] Table 2 outlines various stages of PD.

The main distinguishing feature of PD is the combination of physical and cognitive symptoms. The onset is rapid,

Table 1: Predictors of anxiety caused by an underlying medical issue^[7]

Onset of anxiety symptoms after age of 35 years
Lack of personal or family history of an anxiety disorder
Lack of childhood history of significant anxiety, phobias, or separation anxiety
Lack of avoidance behavior
Absence of significant life events generating or exacerbating the anxiety symptoms
Poor response to antianxiety agents

reaching its peak within 10 min and the attack around 1 h. The typical patient experiences 2–4 attacks per week, commonly accompanied by anticipatory anxiety.^[11]

Panic attack

A brief period of intense fear or discomfort in which ≥ 4 of the following signs or symptoms occur abruptly and peak within 10 min^[12]:

- Palpitations, pounding heart, or accelerated heart rate
- Sweating
- Trembling or shaking
- Sensation of shortness of breath or smothering
- Feeling of choking
- Chest pain or discomfort
- Nausea or abdominal distress
- Feeling dizzy, unsteady, light-headed, or faint
- Derealization (feelings of unreality) or depersonalization (being detached from oneself)
- Fear of losing control or going crazy
- Fear of dying
- Parasthesia (numbness or tingling sensations)
- Chills or hot flushes.

Table 3 outlines the typical and atypical symptoms of panic attacks.

PD: Recurrent, sudden panic attacks that are followed by a minimum of 1 month of persistent concern about having another panic attack, worry about the potential implications or consequences of the attacks, or a significant change in behavior related to the attacks.^[11]

Table 4 outlines the medical conditions that may cause anxiety.

Management of the PD

Assessment

Patients with PD or anxiety usually present to the ED with somatic symptoms of breathlessness, palpitations, chest pain, etc. It is not uncommon for them to be diagnosed as having an acute respiratory or cardiac event or any other physical illness. Even though the patient is having a panic attack, it is highly essential to rule out all acute physical emergencies. A proper history taking along with a careful physical examination of the vital parameters and certain basic investigations help the emergency physician to rule out a physical illness.

These include the measurement of vital data like blood pressure, oxygen saturation, pulse rate and characteristics, and respiratory rate. Hematological investigations like complete blood counts, serum electrolytes, blood glucose levels, arterial blood gas analysis, thyroid function tests, and renal function tests are to be done. Electrolyte abnormalities like reduced ionized calcium and serum phosphate levels are usually seen in patients with hyperventilation. An electrocardiogram to rule out any acute cardiac abnormalities can be done.

Table 2: Stages of panic disorder^[10]

Stage	Name of the stage	Symptomatology
Stages		
I	Limited symptom attack	Patients display fewer than the four symptoms necessary for diagnosis of panic disorder.
II	Panic attack	Patients meet the definition of panic disorders with the appropriate frequency, duration, and four or more of the symptoms.
III	Hypochondrias	Patient becomes preoccupied with concerns about medical illness, despite medical assurances. The panic attacks may become associated with environmental stimuli. This is known as phobic avoidance behavior. Driving and going to stores or shopping malls are the most frequent fears.
IV	Agoraphobia	
V	Extensive phobic avoidance	
VI	Secondary depression	It is believed to result from progressive disability and demoralization.

Table 3: Typical and atypical symptoms of panic attacks^[4]

Typical symptoms	Atypical symptoms
Tachycardia, palpitations	Vice-like/crushing chest pain
“Atypical” chest pain	Pleuritic chest pain
Trembling, shaking	Shaking rigor
Sweating	Diaphoresis (generalized, drenching)
Dyspnea	Stridor
Subjective weakness in arms and legs	Objective muscular weakness
Flushes or chills	Overt lack of coordination
Dry mouth	Fever, generalized erythema, or rash
Choking sensation	Mechanical inability to swallow
Dizziness, lightheaded feeling	True vertigo, syncope
Depersonalization, derealization	Disorientation to person, time, or place
Nausea, abdominal distress	Vomiting (projectile, bilious, or recurrent)
Fear of losing control or other “irrelevant” catastrophe	Bizarre behavior (unrelated to fear of the attack)

Point of care ultrasonography of the lungs, bedside X-ray of the chest, and peak flowmetry to rule out any respiratory abnormalities are advisable. A toxicological screening to rule out any drug abuse is also helpful in an emergency setting. After ruling out the possible physical causes, the patient could be diagnosed with a PD or an anxiety disorder.

Application of DSM-V or ICD-10 guidelines in an emergency setting could be difficult due to the unavailability of a psychiatrist in the ED. This could lead to underdiagnosing or misdiagnosing the PD. Hence, using certain screening tools that can be applied by an emergency physician or any general physician could be useful in diagnosing a patient with panic or anxiety disorders.

Screening tools

Following are some of the available screening tools that can be used by physicians or primary care doctors in an emergency setting [Table 5]:

Table 4: Medical conditions that may cause anxiety^[13]

System	Conditions
Cardiovascular diseases	Congestive heart failure, acute chest pain, acute myocardial infarction, angina, anemia, hypotension, hypertension, arrhythmias, hypovolemia
Respiratory	Asthma, acute and chronic bronchitis, COPD, pneumonia, hyperventilation, sleep apnea
Metabolic syndrome	Hypocalcemia, hypokalemia, porphyria, pellagra, uremia
Endocrine disorders	Hyperadrenocorticism, pituitary dysfunction, hyperthyroidism, hypothyroidism, parathyroid dysfunction, pheochromocytoma, hypoglycemia, virilization disorders in females, premenstrual syndrome
Neurologic disorders	Cerebrovascular disease, cerebral neoplasm, encephalitis, migraines, subarachnoid hemorrhage, closed head injuries, multiple sclerosis, Wilson disease, vestibular dysfunctions, dementia, delirium, Huntington's disease, temporal lobe diseases, seizure disorders, psychomotor epilepsy
Inflammatory disorders	Systemic lupus erythematosus, rheumatoid arthritis, temporal arteritis, fibromyalgia, allergic reactions
Toxicity	Caffeine intoxication, amphetamines, heavy metals, vasopressors and sympathomimetic agents, organophosphates, alcohol, opiates, phencyclidines, cocaine, ecstasy
Infectious and other diseases	Septicemia, carcinoid syndrome, infectious mononucleosis, AIDS, systemic malignancies, subacute bacterial endocarditis, gastrointestinal hemorrhage
Miscellaneous	Irritable bowel syndrome, dyspepsia, GERD, shingles (herpes zoster)

AIDS, acquired immunodeficiency syndrome; COPD, chronic obstructive pulmonary disease; GERD, gastroesophageal reflux disease; HIV, human immunodeficiency virus

Table 5: Screening tools for anxiety/panic disorder

Anxiety Disorder	Panic Disorder
Anxiety Disorder Diagnostic Questionnaire ^[14]	Panic Disorder Self-Report ^[18]
Generalized Anxiety Disorder 7 ^[15]	Panic Disorder Severity Scale ^[19]
Beck Anxiety Inventory ^[16]	Panic and Agoraphobia Scale ^[20]
Hamilton Anxiety Rating Scale ^[17]	NIMH Panic Questionnaire ^[21]
	Panic associated symptoms scale ^[22]

Treatment in an emergency setting

After careful history taking and a careful assessment, patients can be diagnosed with a physical illness or with anxiety or PD. If the patient is found to have a physical illness, he or she can be treated by the emergency physician immediately and later referred to the concerned specialist physician. If the patient is found to have a psychogenic panic attack or an anxiety disorder, he can be treated by the emergency physician to subvert the crisis situation and then referred to a psychiatrist. However, it is not uncommon for patients diagnosed with primary mental illness to have an underlying medical condition, and should be taken into consideration while treating the patient. Patients can be treated using both pharmacological and non-pharmacological measures.

Pharmacotherapy

The most important step in the management is to abort the panic attack. A benzodiazepine with rapid onset of action is the choice of pharmacotherapy. Alprazolam, lorazepam, and clonazepam are used to treat anxiety/panic symptoms. Alprazolam is usually started at a low dose of 0.25 mg, three times a day, but doses of 4 mg/day or more may be required.^[23] The initial starting dose of lorazepam is 2 mg to 3 mg, can repeat the dose 2 to 3 times per day; the maximum dosage is 10 mg per day.^[24] It can also be administered via intravenous (IV) or intramuscular (IM) injection (2 mg/mL solution, and 4 mg/mL solution). The onset of its action is 1 to 3 min if administered IV and 15 to

30 min if administered IM. Clonazepam is given orally at a starting dose of 0.25 mg and can be used with a maximum dose of 4 mg/day. Abrupt cessation of the benzodiazepines might lead to withdrawal and hence careful tapering of the medication must be done. Caution must be taken with the use of long-term benzodiazepines because of the dependence potential.

In the long-term treatment of anxiety/PD, selective serotonin reuptake inhibitors (SSRIs) are the first-line medications. Examples of SSRIs approved by the FDA to treat the PD include escitalopram, paroxetine, sertraline, and fluoxetine. Even though all of them are equally effective, because of their sedating property, paroxetine is more commonly used.^[25] These medications must be titrated slowly and take several weeks to be effective. Common side effects of SSRIs include nausea, diarrhea, constipation, headache, tremors, agitation, dizziness, sweating, and sexual dysfunction. Tricyclic antidepressants include imipramine, clomipramine, etc. Serotonin-norepinephrine reuptake inhibitors (SNRIs) like venlafaxine are also used in the treatment of PD. A brief course of alprazolam can be prescribed in conjunction with an SSRI for short-term management of PD and should be slowly titrated when the therapeutic actions of the SSRI become apparent (2–4 weeks). Medications and their dosages used in PD are summarized in Table 6.

Non-pharmacological therapy

Most panic attacks spontaneously resolve within half an hour, and patients can sometimes present with the attacks subsiding or with feelings of anticipatory anxiety. Therefore, during such an attack, the treating physician should reassure the patient and explain the brief nature of the illness. Psychoeducation of the patient and the caregivers regarding the course and prognosis of the illness helps in alleviating the anxiety. For further information on psychoeducation in the management of anxiety, refer to Clinical Practice Guidelines for Psychoeducation in Psychiatric Disorders

General Principles of Psychoeducation.^[26] In the long term, psychotherapies such as cognitive behavioral therapy,^[27] interpersonal therapy, and mindfulness therapy can be used to treat PD, but also take effect after several weeks. Therefore, healthcare providers in the ED should properly refer and educate on general lifestyle recommendations to reduce and identify any anxiety-related symptoms, such as eliminating stimulants, obtaining adequate sleep, and exercising daily. In addition, relaxation techniques^[17] can easily be administered in ED settings and have the potential to reduce anxiety. Deep breathing exercises,^[28] which involve consciously slowing respirations and focusing on taking regular slow deep breaths have been shown to reduce anxiety or panic symptoms. Another strategy may include guided imagery,^[29] where the emergency physician encourages the patient to imagine a serene location free of stress. Both of these methods may have a profound effect

on the anxiety of patients who present with a panic attack or known PD.

Role of emergency physicians

PD is a very common presentation in the primary care setting, especially in an ED. Most of the patients have cardiac, gastrointestinal, ear, nose, and throat (ENT), or neurological comorbidities. As the first point of contact with the patients, emergency physicians have an important role in the management of PDs. A holistic history and clinical assessment are of paramount importance. Early recognition and management help to reduce both morbidity and mortality. The emergency physician should be able to abate the panic attack and put in a referral to a psychiatrist especially if the patient has suicidality or any self-neglect or inadequate response to the primary intervention. Early referral decreases overall cost in the assessment and management of PDs.^[30] The management of patients presenting to the emergency department has been summarized in Figure 1.

Table 6: Medications used in anxiety/panic disorder

Drug	Initial dose	Maintenance dose
Benzodiazepines		
Alprazolam	0.25-0.5 mg tid	0.5-2 mg tid
Lorazepam	1-2 mg bid	1-2 mg tid
Clonazepam	0.25-0.5 mg bid	0.5-2 mg bid
SSRI		
Paroxetine	5-10 mg	20-60 mg
Fluoxetine	20 mg	20-60 mg
Escitalopram	10 mg	10-20 mg
Sertraline	12.5-25 mg	50-200 mg
TCA		
Clomipramine	5-12.5 mg	50-125 mg
Imipramine	10-25 mg	150-500 mg
Desipramine	10-25 mg	150-200 mg
SNRI		
Venlafaxine	6.25-25 mg	50-150 mg

SUMMARY

Anxiety/PD is common in the general population and often presents to the ED. The emergency physician must be able to identify the underlying cause for the presentation, which could be either a primary mental illness or an underlying medical condition presenting with the symptoms of anxiety/PD to initiate proper treatment. Basic investigations help in the assessment of the symptoms. If the cause is an underlying medical illness, referral to the concerned specialist would be helpful and if symptoms are purely due to a primary mental illness, the patient can be screened for an anxiety or

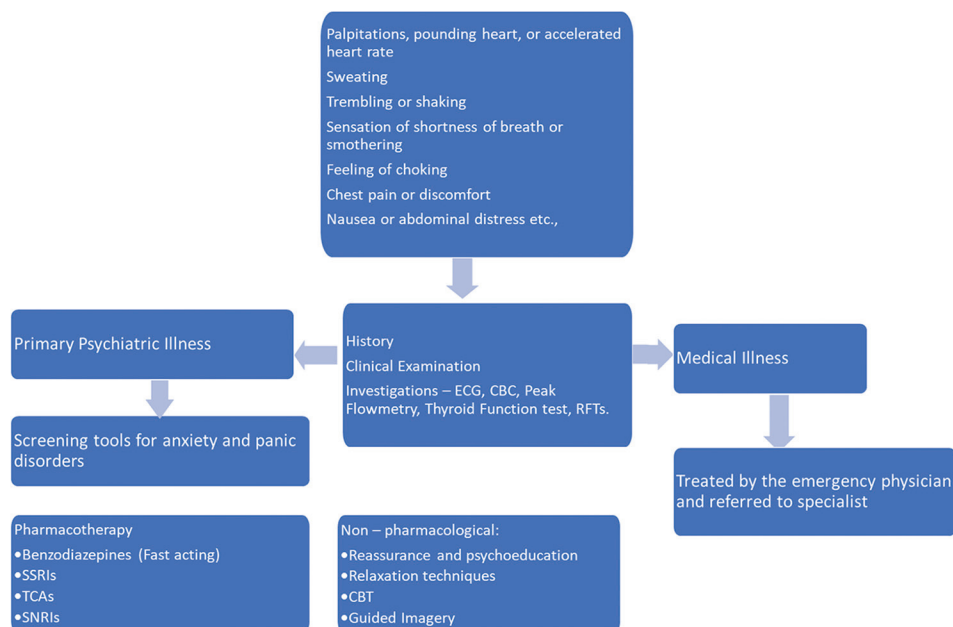


Figure 1: Algorithm depicting the management of patients presenting to the emergency department with anxiety/panic symptoms

PD. Early and timely recognition of primary anxiety disorder helps in avoiding unnecessary investigations and can be treated with both pharmacological and non-pharmacological options. The main aim is to calm down the patient and abort the panic attack. Benzodiazepines and SSRIs remain the mainstay of treatment. Reassurance and psychoeducation of the patient as well as his caregivers help a long way in the management of the symptoms. Judicious use of medication, patient education, and referral to a physician whenever necessary improve the outcome of patients with this otherwise potentially disabling disorder.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

REFERENCES

1. Bandelow B, Michaelis S. Epidemiology of anxiety disorders in the 21st century. *Dialogues Clin Neurosci* 2015;17:327-35.
2. Sandeep K, Rajmeet S. Role of different neurotransmitters in anxiety: A systemic review. *Int J Pharm Sci Res* 2017;8:411-21.
3. Sadock BJ, Sadock VA. Kaplan and Sadock's Concise Textbook of Clinical Psychiatry. 3rd ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2008.
4. Pollard CA, Lewis LM. Managing panic attacks in emergency patients. *J Emerg Med* 1989;7:547-52.
5. Felder ML, Perry MA. The patient with anxiety disorders in the emergency department. In: Zun LS, Chepenik LG, MNS Mallory, editors. *Behavioral Emergencies for the Emergency Physician*. Cambridge: Cambridge University Press; 2013. p. 76-82.
6. Lentz C. Anxiety and mood disorders in an emergency context. *Big Book of Emergency Department Psychiatry*. New York: Productivity Press; 2017. p. 147-76.
7. Zun LS, Nathan JB. Anxiety disorders. *Rosen's Emergency Medicine: Concepts and Clinical Practice*. 7th ed. Philadelphia, PA: Mosby Elsevier; 2009.
8. Merritt TC. Recognition and acute management of patients with panic attacks in the emergency department. *Emerg Med Clin North Am* 2000;18:289-300.
9. Noyes R, Clancy J, Hoenk PR, Slymen DJ. The prognosis of anxiety neurosis. *Arch Gen Psychiatry* 1980;37:173-8.
10. Zun LS. Panic disorder: Diagnosis and treatment in emergency medicine. *Ann Emerg Med* 1997;30:92-6.
11. Manjunatha N, Ram D. Panic disorder in general medical practice-A narrative review. *J Family Med Prim Care* 2022;11:861-9.
12. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders-5*. Arlington, VA: American Psychiatric Publishing; 2013.
13. Tan DT, Khouzam H, Gill T. The Acutely Anxious Patient. In: *Handbook of Emergency Psychiatry*. 1st ed. Philadelphia: Mosby; 2007. p. 214-33.
14. Norton PJ, Robinson CM. Development and evaluation of the anxiety disorder diagnostic questionnaire. *Cogn Behav Ther* 2010;39:137-49.
15. Spitzer RL, Kroenke K, Williams JB, Löwe B. A brief measure for assessing generalized anxiety disorder: The GAD-7. *Arch Intern Med* 2006;166:1092-7.
16. Beck AT, Epstein N, Brown G, Steer RA. An inventory for measuring clinical anxiety: psychometric properties. *J Consulting Clin Psychol* 1988;56:893.
17. Öst LG, Westling BE. Applied relaxation vs cognitive behavior therapy in the treatment of panic disorder. *Behav Res Ther* 1995;33:145-58.
18. Newman MG, Holmes M, Zuellig AR, Kachin KE, Behar E. The reliability and validity of the panic disorder self-report: A new diagnostic screening measure of panic disorder. *Psychol Assess* 2006;18:49-61.
19. Shear MK, Rucci P, Williams J, Frank E, Grochocinski V, Vander Bilt J, *et al.* Reliability and validity of the Panic Disorder Severity Scale: Replication and extension. *J Psychiatr Res* 2001;35:293-6.
20. Bandelow B. *Panic and Agoraphobia Scale (PAS)*. Newbury port, Massachusetts: Hogrefe & Huber Publishers; 1999.
21. Scupi BS, Maser JD, Uhde TW. The National Institute of Mental Health Panic Questionnaire: An instrument for assessing clinical characteristics of panic disorder. *J Nerv Ment Dis* 1992;180:566-72.
22. Argyle N, Deltito J, Allerup P, Maier W, Albus M, Nutzinger D, *et al.* The Panic-Associated Symptom Scale: Measuring the severity of panic disorder. *Acta Psychiatr Scand* 1991;83:20-6.
23. Locke AB, Kirst N, Shultz CG. Diagnosis and management of generalized anxiety disorder and panic disorder in adults. *Am Fam Physician* 2015;91:617-24.
24. Vlastra W, Delewi R, Rohling WJ, Wagenaar TC, Hirsch A, Meesterma MG, *et al.* Premedication to reduce anxiety in patients undergoing coronary angiography and percutaneous coronary intervention. *Open Heart* 2018;5:e000833.
25. Du Y, Du B, Diao Y, Yin Z, Li J, Shu Y, *et al.* Comparative efficacy and acceptability of antidepressants and benzodiazepines for the treatment of panic disorder: A systematic review and network meta-analysis. *Asian J Psychiatr* 2021;60:102664.
26. Sarkhel S, Singh OP, Arora M. Clinical practice guidelines for psychoeducation in psychiatric disorders general principles of psychoeducation. *Indian J Psychiatry* 2020;62(Suppl 2):S319-23.
27. Otto MW, Deveney C. Cognitive-behavioral therapy and the treatment of panic disorder: Efficacy and strategies. *J Clin Psychiatry* 2005;66(Suppl 4):28-32.
28. Taylor S. Breathing retraining in the treatment of panic disorder: Efficacy, caveats and indications. *Scan J Behav Ther* 2001;30:49-56.
29. Öst LG, Westling BE, Hellström K. Applied relaxation, exposure *in vivo* and cognitive methods in the treatment of panic disorder with agoraphobia. *Behav Res Ther* 1993;31:383-94.
30. Chen YH, Chen SF, Lin HC, Lee HC. Healthcare utilization patterns before and after contact with psychiatrist care for panic disorder. *J Affect Disord* 2009;119:172-6.

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

Access this article online	
Website: www.indianjpsychiatry.org	Quick Response Code 
DOI: 10.4103/indianjpsychiatry.indianjpsychiatry_489_22	

How to cite this article: Raju NN, Naga Pavan Kumar KS, Nihal G. Clinical Practice guidelines for assessment and management of anxiety and panic disorders in emergency setting. *Indian J Psychiatry* 2023;65:181-5.